

UPDATED PROJECT PROPOSAL

Title: Enhanced Priority-Based Scheduling and Process Monitoring System in xv6

1. Introduction

This project aims to enhance the xv6 operating system by implementing an advanced priority-based scheduling mechanism along with a process monitoring system. Unlike the default Round Robin scheduler, this project introduces intelligent decision-making in CPU allocation.

2. Problem Statement

The existing xv6 scheduler treats all processes equally, lacking priority differentiation and runtime visibility. This limits its ability to simulate real-world operating system behavior.

3. Objectives

- Implement a priority-based scheduling algorithm
- Maintain fairness using Round Robin for equal priority
- Prevent starvation using aging technique
- Develop system calls for dynamic priority control and monitoring

4. Detailed Implementation Plan

Task 1: Priority-Based Scheduler

Step 1: Modify process structure (proc.h) to include a priority field.

Step 2: Initialize default priority during process creation.

Step 3: Redesign scheduler to select highest priority RUNNABLE process.

Step 4: Integrate Round Robin for same-priority processes.

Step 5: Implement aging to prevent starvation.

Step 6: Add system call `setpriority(pid, value)` for dynamic updates.

Task 2: Process Monitoring System

Step 1: Track runtime, state, and memory usage.

Step 2: Update statistics using timer interrupts.

Step 3: Implement `psinfo()` system call to display process information.

5. Testing and Evaluation

- Verify execution order based on priority
- Validate fairness among same-priority processes
- Ensure starvation is prevented

- Confirm accuracy of monitoring data

6. Challenges and Solutions

- Kernel instability → solved by incremental testing
- Race conditions → handled with synchronization mechanisms
- Incorrect scheduling → debug using kernel logs

7. Timeline

Week 1: Setup xv6 and study scheduler

Week 2: Implement priority scheduler

Week 3: Develop monitoring system

Week 4: Testing and debugging

8. Conclusion

This project enhances xv6 by integrating a hybrid scheduling mechanism that balances efficiency and fairness while providing real-time process insights. It aligns closely with real-world OS design principles.